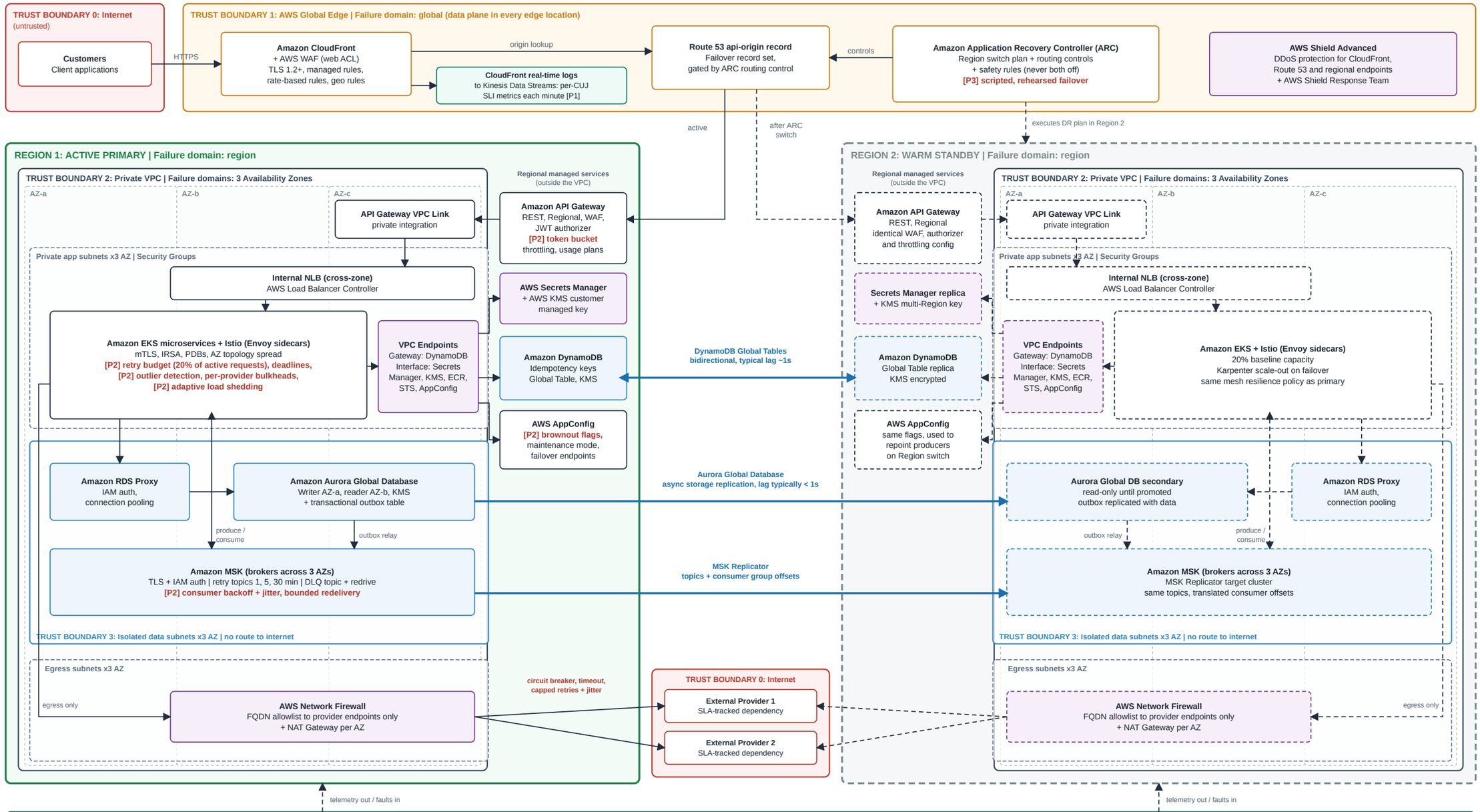
Selected cloud: AWS | Tier 1 Digital Service: Multi-Region Reliability Architecture

Active primary + warm standby | trust boundaries, traffic flow, failure domains, security controls | [P1] to [P4] map controls to the scenario problems



Legend

Live request path · Standby / failover path · Async cross-region replication · Security control · Data store · Availability Zone (failure domain)

Scenario mapping: [P1] slow detection of customer impact [P2] retry storms caused two regional incidents [P3] DR never exercised end to end [P4] unclear reliability ownership

Targets: Customer journey availability 99.99% (error budget about 4.3 min per 30 days, 52.6 min per year) | RTO 45 min via rehearsed ARC Region switch plan | RPO 5 min: async replication typically seconds, alarmed at 60s lag

Design note: AZ failures are absorbed in-region with no failover (the 99.99% budget cannot afford a 45 min regional cutover often). Istio on EKS is used because AWS App Mesh reached end of support on 30 Sep 2026 and Amazon VPC Lattice does not provide retry budgets or outlier detection; external providers sit behind fallbacks so their availability does not cap the journey SLO.